Wigner Resolution from Action Capacity

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Smoothing the Wigner function gives a non-negative phase-space portrait of a quantum state and the scale of that smoothing has been chosen rather than derived. We show that scale is fixed by the state. The state's covariance defines a Heisenberg cell of action capacity $\pi\,\Delta x\,\Delta p$, and the cell admits a family of de Gosson quantum blobs deforming at constant action $h/2$. Convolving the Wigner function to a single $h/2$ blob renders it non-negative. Integrating over the family reaches the resolution $\pi\hbar^2/(\Delta x\,\Delta p)$, the symplectic polar dual of the Heisenberg cell. The portrait at this resolution is everywhere non-negative and holds the structure of the Wigner function in place. Wigner resolution is read from the state's action capacity alone, sharpens as that capacity grows, and carries no free parameter.

The Wigner function represents a quantum state in phase space [1, 2]. It is defined pointwise, below Planck's quantum of action, and Hudson's theorem forces it negative there [3]. Its negativity is a resource for quantum advantage [4, 5] and a measure of nonclassicality [6], while a non-negative distribution supports classical sampling [7–11]. A non-negative portrait follows from smoothing the Wigner function, and the scale of that smoothing has been chosen rather than derived. Here a family of $h/2$ quantum blobs, circumscribed by the state's action capacity $\pi\,\Delta x\,\Delta p$, fixes that scale. The blobs resolve the Wigner function to a non-negative portrait at $\pi\hbar^2/(\Delta x\,\Delta p)$, the finest resolution it reaches, set by the capacity alone with no free parameter.

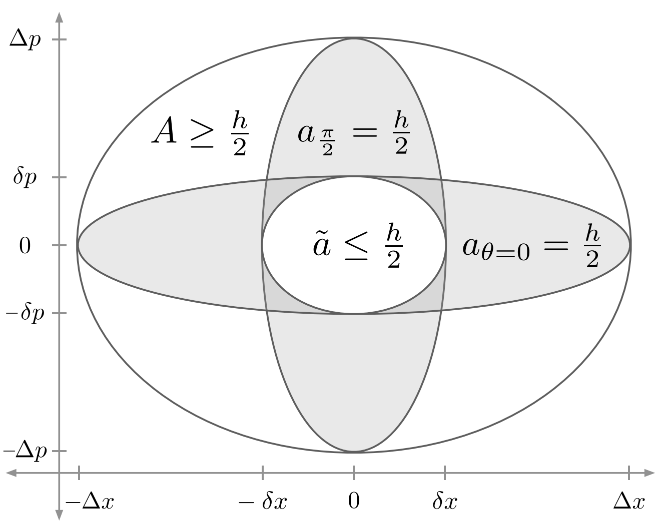

FIG. 1. **Heisenberg cell, principal quantum blobs, and quorum cell.** The Heisenberg cell is the state's covariance ellipsoid with action capacity $A = \pi\,\Delta x\,\Delta p \geq h/2$. The quantum blob family (Fig. 2) has two members at the principal axes, $a_{\theta=0} = \pi\,\Delta x\,\delta p = h/2$ and $a_{\pi/2} = \pi\,\delta x\,\Delta p = h/2$. Zurek's relations $\delta p \sim \hbar/\Delta x$ and $\delta x \sim \hbar/\Delta p$ locate the semi-axis scales from the state's action capacity. The quorum cell $\tilde{a}$ is an ellipse with semi-axes $\delta x, \delta p$ and action $\tilde{a} = \pi\,\delta x\,\delta p$. A circumscribes $a_{\theta=0}$ and $a_{\pi/2}$, which circumscribe $\tilde{a}$.

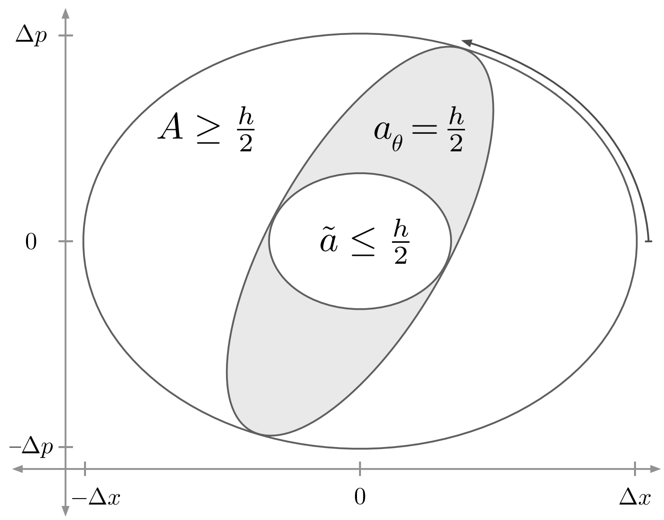

FIG. 2. **Quadrature family of quantum blobs and quorum cell.** The Heisenberg cell A circumscribes the quantum blob family $\{a_\theta\}$, which circumscribes the quorum cell $\tilde{a}$. Across θ the family deforms at constant action $a_\theta = h/2$. At the principal angles $\theta = 0$ and $\theta = \pi/2$, the family recovers $a_{\theta=0}$ and $a_{\pi/2}$ from Fig. 1.

Planck divided phase space into elementary regions of action [12]. Heisenberg's uncertainty principle $\sigma_x\,\sigma_p \geq \hbar/2$ [13–15] fixes the limit. Geometrically, de Gosson gave this action region the form of the state's covariance ellipsoid, with symplectic capacity $\pi\,\Delta x\,\Delta p$ [16] and semi-axes $\Delta x = \sqrt{2}\,\sigma_x$ and $\Delta p = \sqrt{2}\,\sigma_p$. This is the Heisenberg cell of action capacity (Figs. 1, 2),

$$A = \pi\,\Delta x\,\Delta p \geq h/2. \tag{1}$$

The Wigner function carries structure at small scales. Zurek's relations $\delta p \sim \hbar/\Delta x$ and $\delta x \sim \hbar/\Delta p$ locate those scales from the state's action capacity [17]. We identify them as the principal axes of a family of de Gosson quantum blobs (Fig. 1),

$$a_{\theta=0} = \pi\,\Delta x\,\delta p = h/2, \qquad a_{\pi/2} = \pi\,\delta x\,\Delta p = h/2. \tag{2}$$

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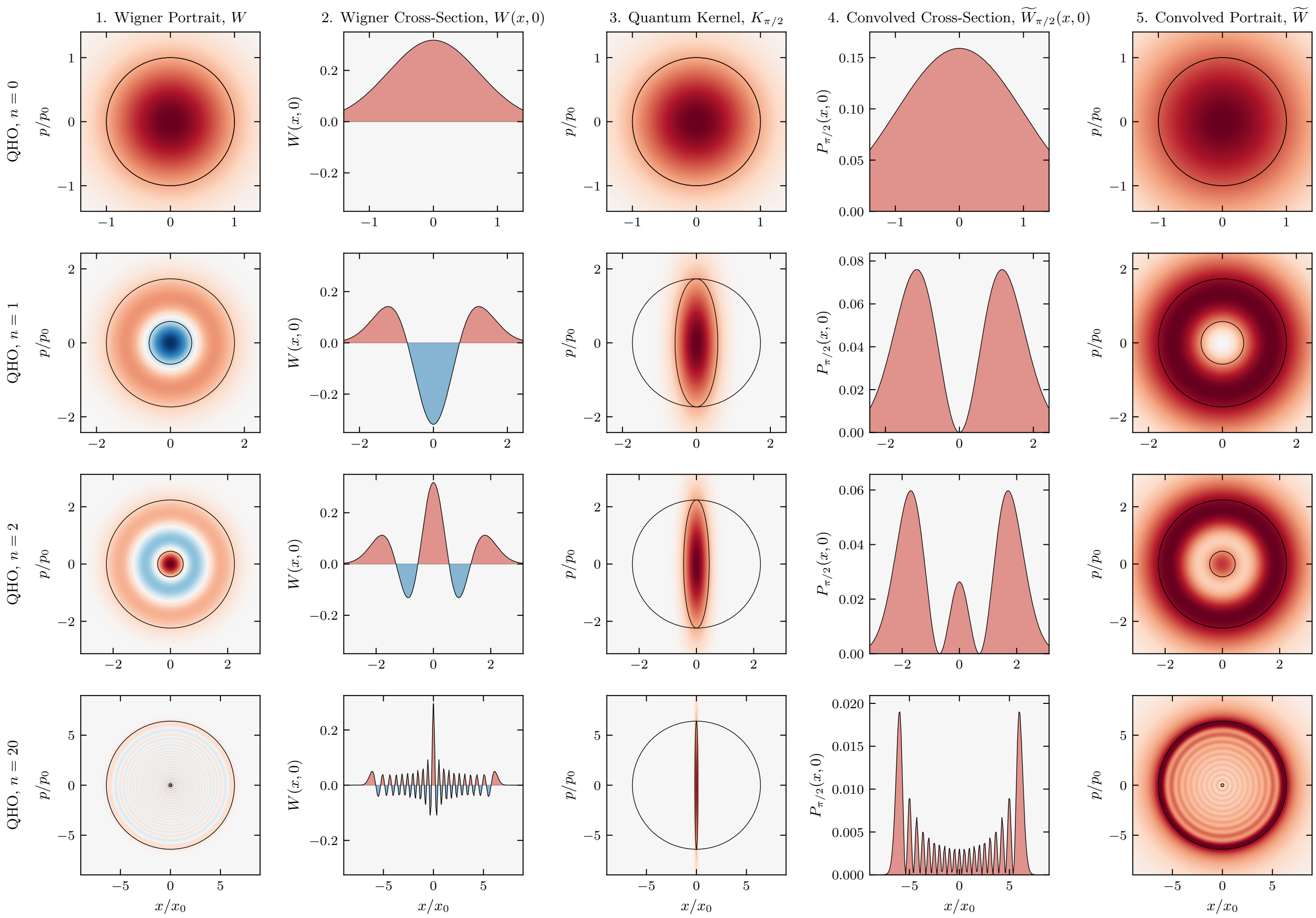

FIG. 3. **Harmonic oscillator.** Rows correspond to Fock states $n = 0, 1, 2, 20$ with action capacity $A/(h/2) = 2n+1$. Column 1: Wigner function $W(x,p)$ (negativity in blue) overlaid with the Heisenberg cell A and quorum cell $\tilde{a}$. Centered at the covariance origin, $\tilde{a}$ marks the resolution of the finest structure (the Zurek scale) of the bare, unconvolved Wigner function. Column 2: Cross-section $W(x,0)$ exhibiting negative dips. Column 3: Quantum kernel $K_{\pi/2}$ overlaid with A and the quantum blob $a_{\pi/2}$ (action $h/2$, restoring resolution up to Planck's quantum of action). Column 4: Convolved cross-section $\widetilde{W}_{\pi/2}(x,0)$ at quadrature angle $\theta = \pi/2$, which is non-negative. Column 5: The portrait $\widetilde{W}(x,p)$, integrated over the blob family across all angles, overlaid with A and $\tilde{a}$ (symplectic polar duals). It is everywhere non-negative, with positive lobes at the positions of those in W, resolved at $\tilde{a}$. For the vacuum ($n = 0$), A and $\tilde{a}$ coincide as a Gaussian blob at $A = h/2$. As n increases, A grows and $\tilde{a}$ reciprocally shrinks ($\tilde{a}\,A = (h/2)^2$). At $n = 20$, the rings condense toward a classical energy trajectory, leaving $\tilde{a}$ as a fine central cell. Axes are in the oscillator scales $x_0 = \sqrt{\hbar/m\omega}$ and $p_0 = \sqrt{\hbar m\omega}$.

The capacity $\Delta x, \Delta p$ runs along the major axes, the resolution $\delta x, \delta p$ along the minor.

Between these principal members the quantum blob family $\{a_\theta\}$ deforms across $\theta \in [0, \pi)$ at constant action $h/2$ (Fig. 2). Each blob a_θ is the preimage of the rigid ellipse $\hat{a}_\theta$ under the scaling $X = x/\Delta x$, $P = p/\Delta p$,

$$a_\theta = \{(x,p) : (X,P) \in \hat{a}_\theta\}\,. \tag{3}$$

The rotated quadratures $X_\theta = X\cos\theta + P\sin\theta$ and $P_\theta = -X\sin\theta + P\cos\theta$ are the homodyne quadratures of optical tomography at phase θ. In the scaled coordinates $\hat{a}_\theta$ is the rotation by θ of a single ellipse,

$$\hat{a}_\theta = \left\{(X,P) : X_\theta^2 + \tfrac{A^2}{(\pi\hbar)^2}\,P_\theta^2 \le 1\right\}, \tag{4}$$

inscribed in the unit disk at every θ, so the Heisenberg cell A circumscribes each blob a_θ.

The Wigner function of the state is

$$W(x,p) = \frac{1}{\pi\hbar}\int \psi^*(x+s)\,\psi(x-s)\,e^{2ips/\hbar}\,ds. \tag{5}$$

The Wigner function of each quantum blob a_θ is the centered Gaussian whose $h/2$ ellipse is a_θ,

$$K_\theta(x,p) = \frac{1}{\pi\hbar}\exp\!\Big[-X_\theta^2 - \tfrac{A^2}{(\pi\hbar)^2}\,P_\theta^2\Big]. \tag{6}$$

The convolution of W with each quantum kernel K_θ is

$$\widetilde{W}_\theta(x,p) = (W * K_\theta)(x,p). \tag{7}$$

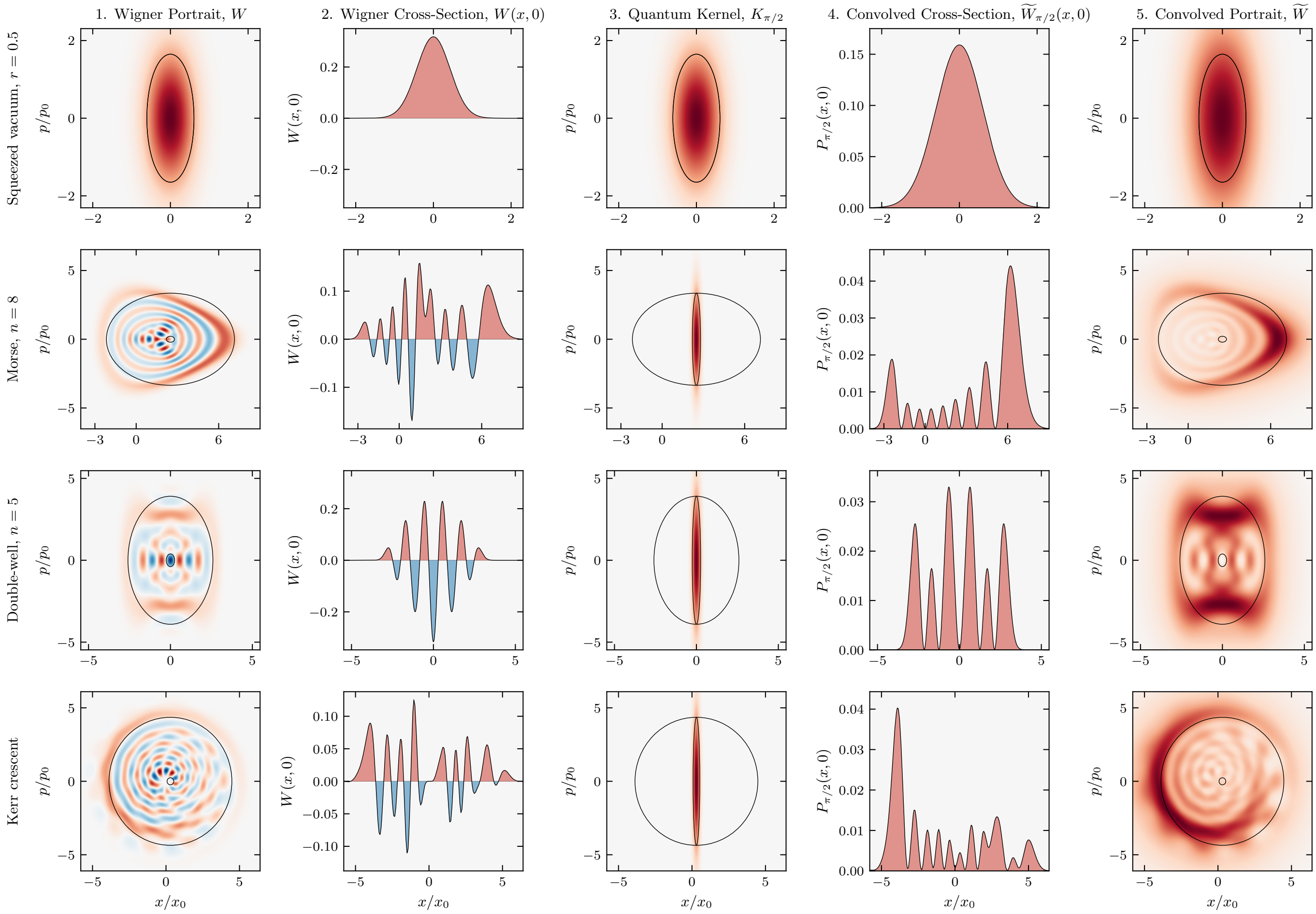

FIG. 4. **Squeezed vacuum, eigenstates, and the Kerr crescent.** Rows correspond to the squeezed vacuum ($r = 0.5$), the Morse eigenstate ($n = 8$), the double-well eigenstate ($n = 5$), and the Kerr crescent, a coherent state sheared at constant photon number. Columns are as in Fig. 3. The Kerr state is shown in the principal frame of its covariance, where the cell overlays are axis-aligned. The portrait $\widetilde{W}$ (Column 5) remains strictly non-negative across all four states, resolved at $\tilde{a}$.

Convolving W with the kernel K_θ of a single quantum blob a_θ lifts it up to Planck's quantum of action and yields a non-negative function [8, 9, 18].

Zurek also located the scale $a \sim \hbar\times(\hbar/A)$ [17, 19] of the chessboard structure in the compass state (Fig. 5, column 1). We identify its rotationally invariant form as the quorum cell $\tilde{a}$. Named after the quorum of observables in optical tomography [20], $\tilde{a}$ is the ellipse with semi-axes $\delta x, \delta p$ (Figs. 1, 2) and action

$$\tilde{a} = \pi\,\delta x\,\delta p = \frac{\pi\hbar^2}{\Delta x\,\Delta p}. \tag{8}$$

$\tilde{a}$ is the resolution of the state's finest structure in the bare Wigner function (Figs. 3, 4, 5, column 1). The quorum cell $\tilde{a}$ is the symplectic polar dual of the Heisenberg cell A [21, 22]. The duality is reflexive, so A and $\tilde{a}$ each fix the other, and antimonotone, so $\tilde{a}$ shrinks as A grows. Each blob a_θ is circumscribed by A and circumscribes $\tilde{a}$, nesting as $A \supseteq a_\theta \supseteq \tilde{a}$ (Fig. 2). A quorum of these a_θ blobs, each respecting $h/2$, reaches $\tilde{a}$ together. Resolution and capacity are reciprocal at the quantum of action, $\tilde{a}\,A = (h/2)^2$.

Integrating $\widetilde{W}_\theta$ over the family gives the phase-space portrait of the state at the resolution $\tilde{a}$,

$$\widetilde{W}(x, p) = \frac{1}{\pi}\int_0^\pi \widetilde{W}_\theta(x, p)\,d\theta. \tag{9}$$

$\widetilde{W}_\theta \geq 0$ for every θ, so $\widetilde{W} \geq 0$ everywhere. Where tomographic reconstruction inverts marginals pointwise to recover W [23–25], convolving W with the family $\{K_\theta\}$ over the same θ builds the non-negative portrait $\widetilde{W}$.

A quantum blob a_θ too large for A to circumscribe, or too small to circumscribe $\tilde{a}$, is a blob of another state and portrays that state, not this one [21, 26]. Because every $h/2$ blob a_θ of this state circumscribes the quorum cell $\tilde{a}$, and a non-negative portrait $\widetilde{W}$ smooths W by a Gaussian of action at least $h/2$ [8, 9], $\tilde{a}$ is the finest resolution any non-negative portrait attains, and the family $\{a_\theta\}$ reaches it across $\theta \in [0, \pi)$.

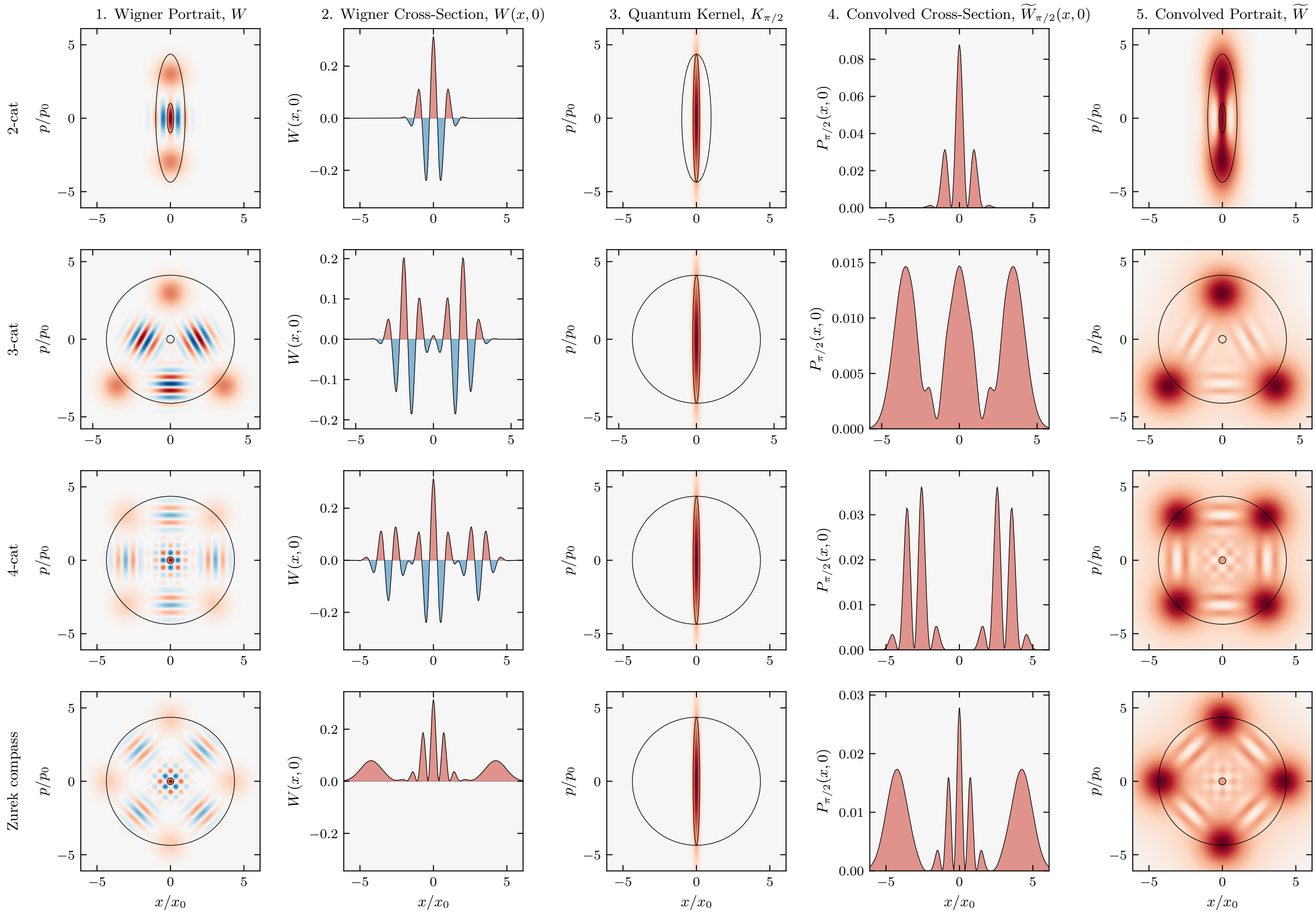

FIG. 5. **Cat states.** Rows correspond to the 2-cat, 3-cat, 4-cat, and Zurek compass states. Columns are as in Fig. 3. As before, the portrait $\widetilde{W}$ (Column 5) is everywhere non-negative, with positive lobes at the positions of those in W, resolved at $\tilde{a}$. The 4-cat and Zurek compass states differ by a $\pi/4$ rotation in phase space.

A fixed vacuum-width Gaussian [7] smooths every state by the same amount, matching only the vacuum $A = h/2$ and erasing the fine structure of every larger state $A > h/2$. Methods with a free parameter [27–29] resolve more finely, the scale chosen rather than derived. The scale here is derived. The state's action capacity A fixes its quorum cell $\tilde{a}$ as the symplectic polar dual, and the family $\{a_\theta\}$ reaches $\tilde{a}$ with no quantity left free.

The principal frame, where the Heisenberg cell A is axis-aligned, costs no generality. The Wigner function transforms covariantly under the symplectic group [18], so the map that diagonalizes a state's covariance carries it to this frame at preserved action. The Kerr crescent, a sheared coherent state, is resolved this way (Fig. 4). In one degree of freedom, A fixes the family uniquely. In higher dimensions, action $h/2$ no longer singles out a blob [26], symplectic capacity and volume diverge, and the higher-dimensional case remains for future work.

Figures 3, 4, and 5 show numerical results across twelve states. The cross-spectral phase between W and $\widetilde{W}$ vanishes, so the portrait holds every feature in place, and $\widetilde{W}$ is non-negative to machine precision [30].

The uncertainty product $\sigma_x\sigma_p$ grows unbounded, the micro-to-macro transition Heisenberg [31] and Bohr [32] identified as requiring special care. The quantum blob does not. As A grows, the family $\{a_\theta\}$ deforms at fixed action $h/2$. Squeezing reaches the same resolution by external apparatus [33], while here the deformation is set by the state. A Schrödinger cat with capacity $\Delta x \sim 10$ cm and $\Delta p \sim 1$ kg·m/s carries structure in its Wigner function at $\delta x = \hbar/\Delta p \sim 10^{-34}$ m [34]. The correspondence principle follows from $A \to \infty$, the physical limit underlying the heuristic $h \to 0$.

The scale called sub-Planck [17] is the minor axis δ of an $h/2$ blob, narrowing as the conjugate capacity Δ grows. The negative Wigner function W and its non-negative portrait $\widetilde{W}$ have their structure at a single resolution $\tilde{a}$, set by the state's action capacity A. Heisenberg's product gives the capacity. The Zurek scales are the axes of de Gosson quantum blobs, and give the resolution. As the action capacity grows to $\pi\,\Delta x\,\Delta p \geq h/2$ the resolution sharpens to $\pi\hbar^2/(\Delta x\,\Delta p) \leq h/2$. We read this structure as the geometry of phase space itself, which de Gosson takes as prior to the uncertainty principle [35].

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